THE NATIONAL UNIVERSITY OF LESOTHO

COURSE OUTLINE			
Course Title	Ordinary Differential Equations		
Course Code	M3503	**No. of Credits**	14
Department	MACS	**Faculty**	Science and Technology
Pre-requisites Course Code	M2502	**Co-requisites Course Code**	None
Course Facilitator(s)	Mosito Lekhooana (Mr)		
Email(s)	mosito.lekhooana@gmail.com / m.lekhooana@nul.ls	**Telephone No.**	5221 3441 Ext. 3441
Other Course Teacher(s) / Tutor(s)	None		
Notional Learning Hours	Total: 140 Learning Hours Pre-reading Materials (20 hrs), Lectures (24 hrs), Class Exercises, Case Studies and/or Computer Simulations (26 hrs), Private Study (40 hrs), and Quiz, Assignment & Exam (30 hrs)		
Student Quota	70		
Course Type	■ Core Course ☐ Compulsory ☐ Elective ☐ Other.....		
Offer in Academic Year	■ 1st Semester ☐ 2nd Semester ☐ Year Long		

COURSE PURPOSE/OVERVIEW

Introduction to the theory of differential equations and their applications. Includes analytical techniques for solving differential equations, the structure of solutions, the existence and uniqueness of solutions, and qualitative behaviour of solutions. Taught for majors in mathematics, science, and engineering.

COURSE CONTENT

Unit 1: First-Order Ordinary Differential Equations

- Revision of
    - Separable Equations
    - Linear Equations
    - Exact Equations
    - Homogeneous Equations
    - Reducible Ordinary Differential Equations (e.g. Bernoulli, Riccati)

Unit 2: Higher Order Linear Differential Equations

- Second-order ODE (homogeneous and non-homogeneous) or higher-order ODE
- Reduction of Order Method
- Homogeneous Linear Equations with Constant Coefficients
- Undetermined Coefficients Method
- Variation of Parameters Method
- Linear differential operators (Inverse Method)

Unit 3: Laplace Transforms

- Definition and properties of the Laplace Transform
- Inverse Laplace Transform
- First and Second Shift Theorems
- Solving Initial Value Ordinary Differential Equations

Unit 4: Linear Systems of Differential Equations

- Constant coefficient homogeneous linear systems
    - Elimination method
    - Eigenvalue problems
- Undetermined coefficients for non-homogeneous linear systems
- Variation of parameters for non-homogeneous linear systems

Unit 5: Power series solutions

- Power series Solutions about ordinary points
- Series solutions about regular singular points
- Series solutions for a point at infinity

COURSE DELIVERY MODE(S) – FACE-TO-FACE TEACHING & LEARNING

1. Face-to-face interactive 1-hour sessions (3 times a week) plus a two-hour tutorial on campus
2. Working on group/individual assignments (4 assignments in a semester)
3. Sitting for a 3 hrs written examination at the end of the semester

COURSE EXIT LEARNING OUTCOMES (ELOs)

Hard Skills ELOs (Subject-Specific Competencies): After the course completion, students will be able to:

#H1	Analyse and solve ordinary initial and boundary value problems of ordinary differential equations
#H2	Compare different analytical methods used to solve ordinary differential equations
#H3	Solve systems of ordinary differential equations
#H4	Use series methods to construct solutions of linear differential equations and initial-/boundary-value problems
#H5	Apply concepts of ordinary differential equations and multivariate calculus to various applications, including real-life problems

Soft Skills ELOs (Generic Competencies): After the course completion, students will be able to:

#S1	Problem-solving skills: the ability to engage with unfamiliar problems and identify relevant solution strategies
#S2	Analytical skills: the ability to construct and express logical arguments and to work in abstract or general terms to increase the clarity and efficiency of analysis
#S3	Collaborative skills: the ability to work in a team [LQF-KK#1, LQF-AC#2]
#S4	Time-management skills: the ability to meet regular deadlines while balancing competing commitments
#S5	Produce a technical report (Writing Skills); [LQF-KK#1, LQF-KK#3, LQF-NS#3, LQF-NS#4, LQF-AC#1, LQF-AC#4, LQF-AC#5]

The information in (round brackets) shows the Programme Competencies with which each outcome aligns, while the one in [square brackets] illustrates alignment to the appropriate Lesotho Qualifications Framework (LQF)'s Level 9 Descriptors (KK – Kinds of Knowledge; NS – Nature of Skills; AC – Agency and Context).

TEACHING AND LEARNING ACTIVITIES

The students will learn the hard and soft skills by:

Attending lectures and reading class notes/presentation slides (H1 to H6)
Engaging in-class group work (H1 to H6, S3 to S5)
Undertaking an individual assignment (H1 to H6, S1, S2, S5, S6)
Preparing for and writing a final examination (H1 to H6, S1 to S5)

COURSE ASSESSMENT METHODS

The student will demonstrate achievement of the Exit Learning Outcomes by:

Assessment Method	**% of assessment towards course grade**
Individual/group work on assignments and quizzes (H1 to H6, S3 to S5)	10
Writing two tests (H1 to H6, S1, S2, S5, S6)	23⅓
Writing the final examination (H1 to H6)	66⅔

ESSENTIAL READINGS

Textbooks:

1. Nagle R. K., Saff E. B. and Snider A. D., Fundamentals of Differential Equations, Addison-Wesley 8th Ed., 2012.
2. Zill D. G., A First Course in Differential Equations with Modelling Applications, Thomson (Brooks/Cole), 2005
3. Hermann M. and Saravi M., A First Course in Ordinary Differential Equations: Analytic and Numerical Methods, Springer, 2014
4. Jeffrey A., Linear Algebra and Ordinary Differential Equations, Blackwell Scientific Publications, 1990
5. Rainville E. D., Bedient P. E. and Bedient R. E., Elementary Differential Equations, Prentice Hall, 1997

Learning Management System (LMS):

All the applicable material, including electronic lectures, etc., will be found in the M3503 Resources Folder on the Thuto LMS platform (https://thuto.nul.ls/)

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

You will get feedback as soon as marking has been completed, approximately within three weeks.

COURSE POLICY (including plagiarism, academic honesty, attendance, etc.)
Unless otherwise stated, the normal course administration policies and rules of the Faculty of Science and Technology apply. For the policy, see http://www.nul.ls/technology/regulations/ All students' submissions are run through Turnitin (https://www.turnitin.com), the anti-plagiarism software that the university has subscribed to. The University Regulations on academic misconduct will be strictly enforced. General health and safety information can be obtained from http://www.nul.ls/services/

ADDITIONAL COURSE INFORMATION
Check the **Khan Academy** and any other **credible** online sources. Try to download some relevant e-books where possible.